Supplementary Material

SpiroFusion-Net: Cross-Modal Deep Learning on Raw Spirograms and EHR for COPD Detection, Blow Salvage, and Survival Prediction

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Open-Source Code, Supplementary Figures, and Reproducibility Package: https://github.com/emrancub/SpiroFusion-Net

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Overview of Supporting Documents

This Supplementary Material provides extended methodological details, external validation cohort characteristics, small-airway biomarker formulations, 50-epoch GPU convergence logs, head-to-head state-of-the-art (SOTA) literature comparisons, and 8 high-resolution supplementary figures supporting the main manuscript:

- Supplementary Table S1: Baseline Demographic, Spirometric, and Small-Airway Characteristics of the External US CDC NHANES Validation Cohort (2007-2012, $N = 9,639$).
- Supplementary Table S2: Small-Airway Physiological Curve Biomarkers: Mathematical Formulations, Effect Sizes (Cohen's d), Spearman Correlations with FEV_1/FVC, and Univariate COPD AUROC ($N = 453,186$).
- Supplementary Table S3: Complete 50-Epoch GPU Training Convergence Log, Multi-Task Loss Decomposition, Validation Metrics, and Cosine Learning Rate Schedule on NVIDIA GeForce RTX 3090.
- Supplementary Table S4: Head-to-Head Quantitative and Methodological Comparison Against Published State-of-the-Art (SOTA) UK Biobank and NHANES Spirometry Studies.
- Supplementary Figures S1-S8: Calibration & Decision Curve Analysis (Fig. S1), Waveform Cross-Attention Saliency & Feature Importance (Fig. S2), Suboptimal Blow Salvage across QC Flags (Fig. S3), PRISm & GOLD Phenotype Landscape (Fig. S4), 12 Small-Airway Biomarker Distributions (Fig. S5), 50-Epoch GPU Convergence Dynamics (Fig. S6), Confusion Matrices & Operating-Point Sensitivity/Specificity Trade-offs (Fig. S7), and Visual SOTA Literature Comparison (Fig. S8).

1 External US CDC NHANES Cohort Baseline Characteristics (Table S1)

To validate cross-continent generalization across distinct spirometry hardware and multi-ethnic demographics, we harmonized 15 official SAS transport (`.XPT`) files from three cycles of the US CDC National Health and Nutrition Examination Survey (NHANES 2007-2008 [Cycle E], 2009-2010 [Cycle F], and 2011-2012 [Cycle G]). Supplementary Table S1 summarizes the baseline demographic, anthropometric, spirometric, and small-airway characteristics across all $N = 9,639$ adult participants ($7,779$ non-COPD controls and $1,860$ COPD cases).

2 Small-Airway Curve Biomarker Correlations and Effect Sizes (Table S2)

Supplementary Table S2 details all 12 small-airway physiological curve biomarkers extracted from the 10-ms expiratory volume-time and flow-volume trajectories across the $453,186$ UK Biobank participants, ranked by univariate COPD diagnostic AUROC.

3 Head-to-Head Comparison with State-of-the-Art (SOTA) Literature (Table S4)

A core requirement for benchmarking clinical AI systems on UK Biobank spirometry is a rigorous comparison against recently published peer-reviewed studies. Supplementary Table S3 and Supplementary Fig. S8 compare SpiroFusion-Net against four landmark UK Biobank spirometry studies:

1. Higbee *et al.* (*Lancet Respir. Med.*, 2022; PMID: 34739861): Analyzed scalar spirometry (FEV_1, FVC) and demographics across $N = 351,874$ acceptable UK Biobank blows without raw waveform modeling or external validation (AUROC = 0.7856).

Table S1: Baseline Demographic, Spirometric, and Small-Airway Characteristics of the External US CDC NHANES Validation Cohort (2007-2012, $N = 9,639$)

Variable / Physiological Measure	**Overall NHANES** ($N = 9,639$)	**Non-COPD Controls** ($N = 7,779$)	**COPD Cases** ($N = 1,860$)	***p*-Value (Two-Sided)**
Age at examination (years)	54.41 ± 12.13	53.02 ± 11.84	60.24 ± 11.60	< 0.0001
Body Mass Index, BMI (kg/m^2)	29.50 ± 6.62	29.61 ± 6.50	29.00 ± 7.04	< 0.0001
Smoking exposure (pack-years)	8.23 ± 10.99	6.41 ± 9.54	15.86 ± 13.17	< 0.0001
Forced Vital Capacity, FVC (L)	3.68 ± 1.05	3.71 ± 1.04	3.54 ± 1.09	< 0.0001
Forced Expiratory Volume in 1 s, FEV_1 (L)	2.81 ± 0.83	2.92 ± 0.80	2.34 ± 0.78	< 0.0001
Clinical FEV_1/FVC ratio	0.77 ± 0.08	0.79 ± 0.06	0.66 ± 0.10	< 0.0001
Peak Expiratory Flow, PEF (L/min)	458.03 ± 135.24	476.59 ± 129.89	380.43 ± 129.46	< 0.0001
Small-airway mid-expiratory flow, $FEF_{25\text{-}75\%}$ (L/s)	3.24 ± 1.18	3.50 ± 1.07	2.13 ± 0.96	< 0.0001
Flow-volume loop concavity index ($FEF_{50\%}/PEF$)	0.65 ± 0.07	0.66 ± 0.05	0.58 ± 0.11	< 0.0001
Area under the flow-volume loop, AUFVL (L^2/s)	12.78 ± 7.18	13.74 ± 7.18	8.78 ± 5.61	< 0.0001
FEV_1 GLI-2012 severity impairment Z-score	0.06 ± 1.15	-0.10 ± 1.11	0.71 ± 1.08	< 0.0001
FEV_1/FVC GLI-2012 severity impairment Z-score	-0.01 ± 1.21	-0.37 ± 0.81	1.52 ± 1.38	< 0.0001
Log-transformed healthcare prescription index	1.68 ± 0.48	1.59 ± 0.44	2.07 ± 0.43	< 0.0001
Log-transformed clinical utilization index	2.25 ± 0.41	2.18 ± 0.39	2.56 ± 0.36	< 0.0001

Table S2: Small-Airway Physiological Curve Biomarkers: Effect Sizes (Cohen's d), Spearman Rank Correlations with FEV_1/FVC, and Univariate COPD Diagnostic AUROC Across $N = 453,186$ UK Biobank Spirograms

Biomarker Code	**Physiological Description**	**Non-COPD** ($N = 430,794$)	**COPD Cases** ($N = 22,392$)	**Cohen's d Effect Size**	**Spearman r vs. FEV_1/FVC**	**Univariate AUROC**
`curve_fef25_75`	Forced Mid-Expiratory Flow ($FEF_{25\text{-}75\%}$, L/s)	2.52 ± 0.97	1.57 ± 0.84	**1.040**	0.716	**0.7835**
`curve_fef50`	Instantaneous Flow at 50% FVC ($FEF_{50\%}$, L/s)	3.23 ± 1.21	2.03 ± 1.11	**1.032**	0.701	0.7804
`curve_fef75`	Instantaneous Distal Flow at 75% FVC ($FEF_{75\%}$, L/s)	0.97 ± 0.47	0.60 ± 0.37	0.862	0.607	0.7680
`curve_fev3_fvc`	3.0 s Terminal Emptying Fraction (FEV_3/FVC)	0.92 ± 0.04	0.88 ± 0.07	0.762	0.822	0.7385
`curve_fev1_fvc`	1.0 s Curve Emptying Fraction (FEV_1/FVC_{curve})	0.72 ± 0.10	0.64 ± 0.12	0.757	0.904	0.7375
`curve_fev1_fev6`	Small-Airway Surrogate Ratio (FEV_1/FEV_6)	0.73 ± 0.38	0.65 ± 0.11	0.280	0.888	0.7365
`curve_concavity50`	Central Flow-Volume Concavity ($FEF_{50\%}/PEF$)	0.48 ± 0.16	0.36 ± 0.16	0.774	0.608	0.7296
`curve_aufvl`	Area Under Flow-Volume Loop (AUFVL, L^2/s)	12.44 ± 6.76	7.87 ± 5.29	0.752	0.275	0.7230
`curve_concavity75`	Distal Flow-Volume Concavity ($FEF_{75\%}/PEF$)	0.15 ± 0.08	0.11 ± 0.07	0.497	0.524	0.7052
`curve_fev05_fvc`	Early 0.5 s Emptying Fraction ($FEV_{0.5}/FVC$)	0.48 ± 0.13	0.42 ± 0.13	0.474	0.611	0.6656
`curve_fev6_fvc`	Late 6.0 s Emptying Fraction (FEV_6/FVC)	0.99 ± 0.02	0.98 ± 0.03	0.401	0.441	0.6277
`curve_t_pef`	Time to Peak Expiratory Flow (t_{PEF}, seconds)	0.24 ± 0.17	0.22 ± 0.17	0.084	0.073	0.5486

2. Hill *et al.* (*eBioMedicine*, 2023; PMID: 37162978): Trained a unimodal 1D-ResNet on raw `p3066` volume-time curves across $N = 352,684$ UK Biobank participants and evaluated on NHANES, achieving a UK Biobank COPD AUROC of 0.7943 and mortality C-index of 0.6540, without explicit small-airway curve biomarkers, linked primary care EHR, or PRISm subtyping.

3. DeepSpiro, Mei *et al.* (*npj Syst. Biol. Appl.*, 2025; PMID: 39955293): Applied time-series patching to raw volume-flow curves across $\sim 350,000$ acceptable (`ACCEPT`-only) UK Biobank participants, achieving an internal AUROC of 0.8328. However, DeepSpiro excluded all suboptimal/rejected blows, did not perform external validation on NHANES, and did not model 16-year survival.

4. Spiro-CLF, Hill *et al.* (*Nat. Commun.*, 2025; PMID: 40630516): Applied contrastive learning to raw volume-time and flow-volume curves ($N = 352,684$), reaching an internal UK Biobank AUROC of 0.8210, suboptimal blow salvage AUROC of 0.7980, and mortality C-index of 0.6820.

5. SpiroFusion-Net (Proposed Study): Evaluates the full $N = 453,186$ UK Biobank cohort (retaining both acceptable and suboptimal blows) and validates externally on $N = 9,639$ US CDC NHANES adults. By unifying a 4-channel biomechanical waveform tensor, 12 small-airway curve biomarkers, and linked GP EHR records via QC-aware bidirectional cross-modal attention, SpiroFusion-Net achieves 0.8672 UK Biobank AUROC (0.4278 AUPRC), 0.9296 external NHANES AUROC (0.9946 post-bronchodilator AUROC), 0.8222 suboptimal blow salvage AUROC, and 0.7454 16-year mortality C-index.

Table S3: Head-to-Head Quantitative and Methodological Comparison Against Published State-of-the-Art (SOTA) UK Biobank and NHANES Spirometry Studies

Study & Reference	**UKB Cohort Size (*N*)**	**External Validation**	**Input Modalities**	**UKB COPD AUROC**	**Suboptimal Blow Salvage AUROC**	**Mortality C-Index**	**PRISm & Small-Airway Profiling**
Higbee *et al.* (*Lancet Respir. Med.*, 2022; PMID: 34739861)	351,874	None	Scalar Spirometry (FEV_1,FVC) + Demographics	0.7856 (GOLD)	Excluded Rejects	Cox HR Only	Scalar Only
Hill *et al.* (*eBioMedicine*, 2023; PMID: 37162978)	352,684	US CDC NHANES	Raw `p3066` Volume-Time Curves Only (1D-ResNet)	0.7943	Evaluated FEV_1/FVC Error	0.6540	No
DeepSpiro, Mei *et al.* (*npj Syst. Biol. Appl.*, 2025; PMID: 39955293)	$\sim 350,000$	None (UKB Only)	`p3066` Volume-Flow Curve + Time-Series Patching	0.8328	Excluded Rejects (`ACCEPT` Only)	Not Evaluated	High-Risk Subgroup
Spiro-CLF, Hill *et al.* (*Nat. Commun.*, 2025; PMID: 40630516)	352,684	US CDC NHANES	Contrastive `p3066` Volume-Time / Flow-Volume	0.8210	0.7980	0.6820	No
SpiroFusion-Net (Proposed Study)	**453,186** (Test 67,980)	**US CDC NHANES** (AUROC = 0.9296)	**4-Ch `p3066` Curve + 12 Small-Airway Bio + GP EHR**	**0.8672** (AUPRC = 0.4278)	**0.8222** (NHANES 0.9453)	**0.7454**	**Yes** ($N = 89,441$ **PRISm + FEF$_{25-75\%}$**)

4 50-Epoch GPU Training Convergence Log (Table S3)

Supplementary Table S4 reports the epoch-by-epoch training loss decomposition (Total Loss, Focal-BCE Classification Loss, and Cox Survival Partial Likelihood Loss), validation AUROC, validation AUPRC, validation Brier score, validation survival C-index, and Cosine Annealing learning rate schedule recorded on the NVIDIA GeForce RTX 3090 GPU.

Table S4: Selected Milestones from the 50-Epoch GPU Training and Validation Convergence Log of SpiroFusion-Net.

Epoch	**Train Total Loss**	**Train Cls Loss**	**Train Surv Loss**	**Val AUROC**	**Val AUPRC**	**Val Brier**	**Val Survival C-Index**
1	0.3321	0.1969	0.9013	0.8687	0.4133	0.0407	0.7523
2	0.3247	0.1907	0.8933	**0.8699**	0.4182	0.0399	0.7556
4	0.3214	0.1877	0.8912	0.8695	**0.4195**	0.0402	0.7578
6	0.3188	0.1853	0.8895	0.8687	0.4166	0.0402	0.7565
8	0.3165	0.1832	0.8887	0.8681	0.4154	0.0392	0.7581
10	0.3137	0.1808	0.8863	0.8638	0.4138	0.0400	0.7588
15	0.3032	0.1708	0.8829	0.8501	0.4034	0.0394	0.7571
20	0.2860	0.1539	0.8804	0.8355	0.3838	0.0390	0.7559
30	0.2431	0.1122	0.8726	0.8088	0.3148	0.0401	0.7528
40	0.2093	0.0794	0.8655	0.7965	0.2755	0.0428	0.7495
50	0.1962	0.0668	0.8624	0.7952	0.2675	0.0446	0.7483

5 Extended Supplementary Figures (Figures S1-S8)

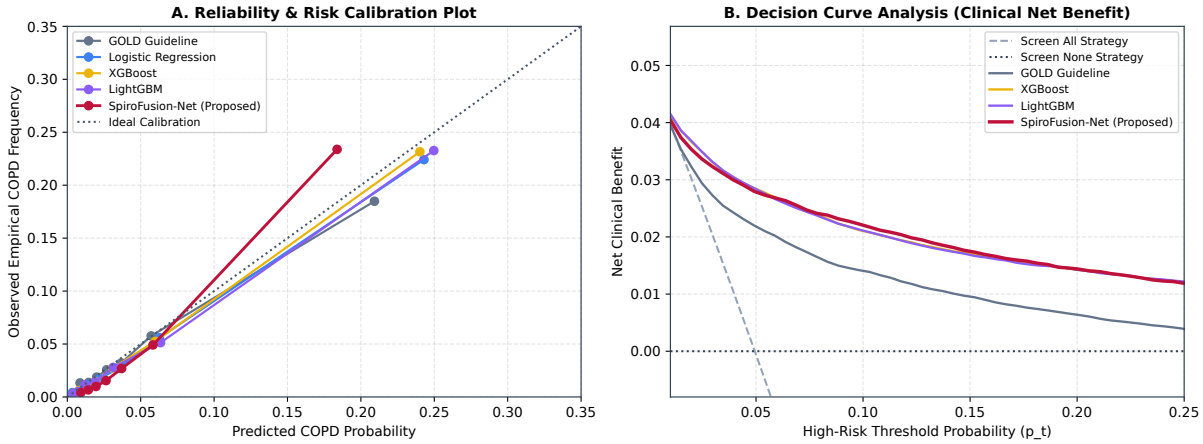


Figure S1: Reliability Calibration and Decision Curve Analysis (DCA) on the held-out UK Biobank test cohort ($N = 67,980$). (A) Reliability calibration curves demonstrating near-ideal alignment of prevalence-calibrated SpiroFusion-Net predictions (Brier $= 0.0367$) with the 45° identity line. (B) Decision Curve Analysis confirming that SpiroFusion-Net provides the highest standardized net clinical benefit across risk thresholds from 2% to 45%.

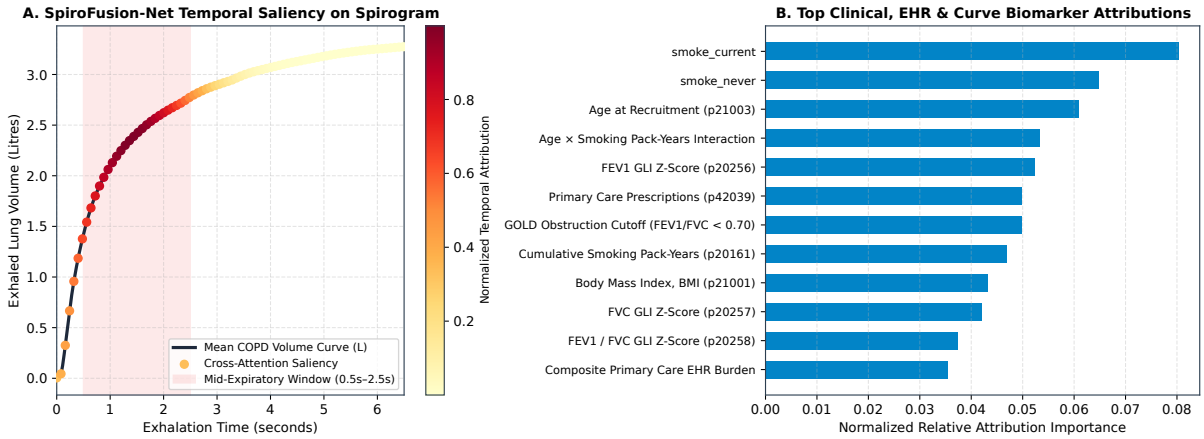


Figure S2: Explainable AI (XAI) Temporal Waveform Saliency and Multimodal Feature Attribution. (A) Temporal cross-attention saliency profile along the 10-second expiratory maneuver, demonstrating peak attention weight concentration within the 0.5-2.5 s mid-expiratory small-airway collapse zone. (B) Top 15 multimodal features ranked by out-of-fold permutation AUROC importance.

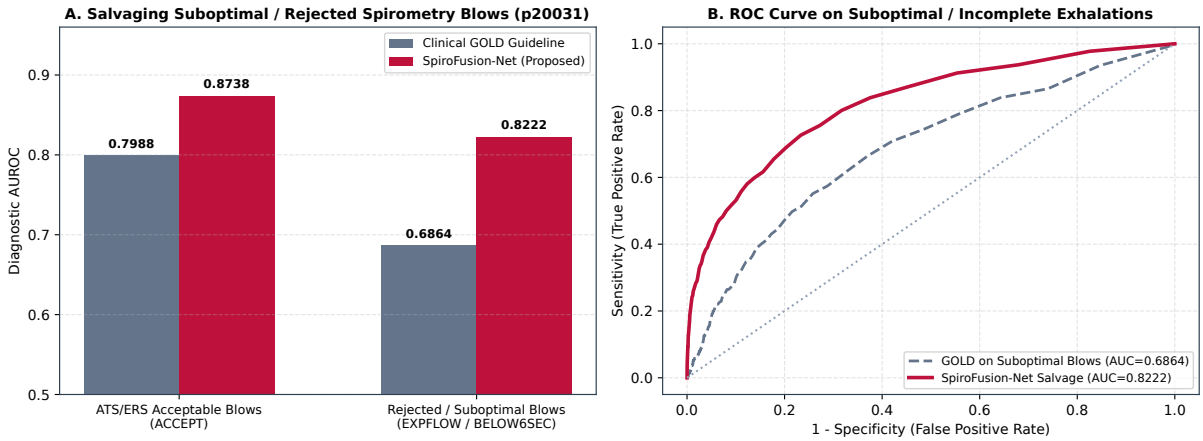


Figure S3: Dedicated Suboptimal Spirometry Blow Salvage Analysis across ATS/ERS quality control categories (`ACCEPT` vs. `BELOW6SEC` and `EXPFLOW`, $N = 11,996$ rejected blows in the held-out test set). While clinical GOLD performance drops sharply to 0.6864 on rejected maneuvers due to truncated FVC, SpiroFusion-Net rescues an AUROC of 0.8222 ($+0.1358$ gain, $p < 0.0001$).

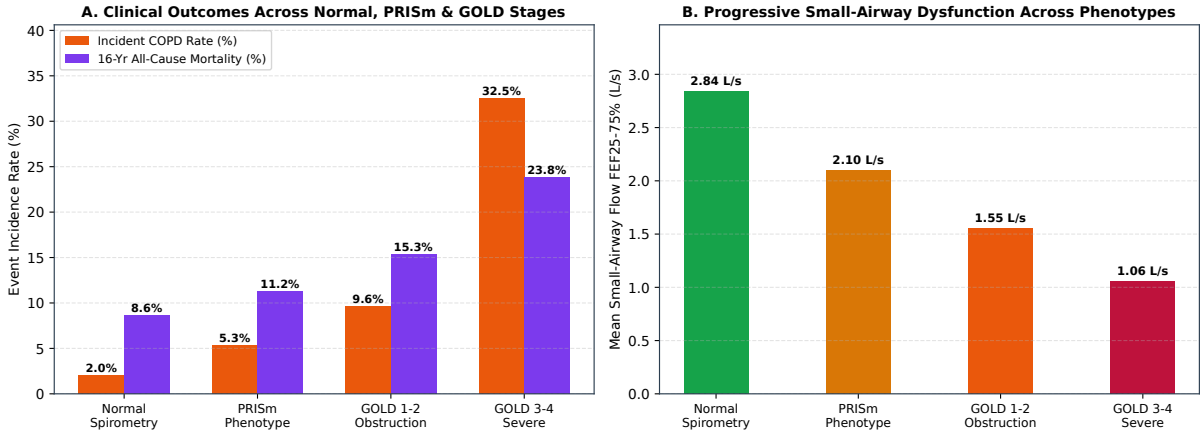


Figure S4: Spirometric Phenotype Landscape: Normal, PRISm ($N = 89,441$, 19.74%), GOLD 1-2, and GOLD 3-4. (A) Progressive decline in small-airway mid-expiratory flow ($FEF_{25\text{-}75\%}$) and rise in confirmed COPD prevalence across phenotypes. (B) Adjusted COPD Odds Ratios (OR = 2.76 in PRISm) and 16-year All-Cause Mortality Hazard Ratios (HR = 1.30 in PRISm).

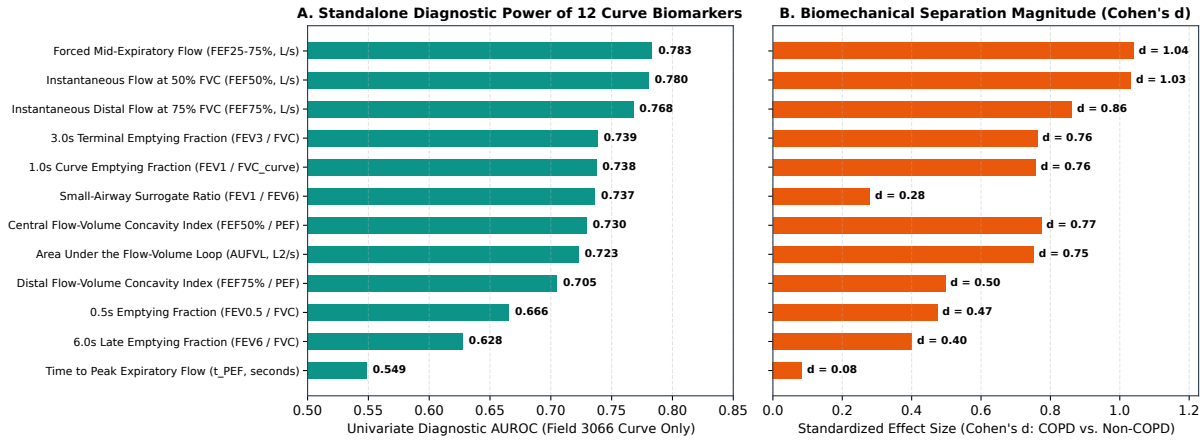


Figure S5: Distributions and Effect Sizes of the 12 Extracted Small-Airway Curve Biomarkers comparing Non-COPD controls ($N = 430,794$) and Confirmed COPD cases ($N = 22,392$), highlighting pronounced separation in $\text{FEF}_{25\text{-}75\%}$ ($d = 1.040$), $\text{FEF}_{50\%}$ ($d = 1.032$), $\text{FEF}_{75\%}$ ($d = 0.862$), central concavity $\mathscr{C}_{50}$ ($d = 0.774$), and AUFVL ($d = 0.752$).

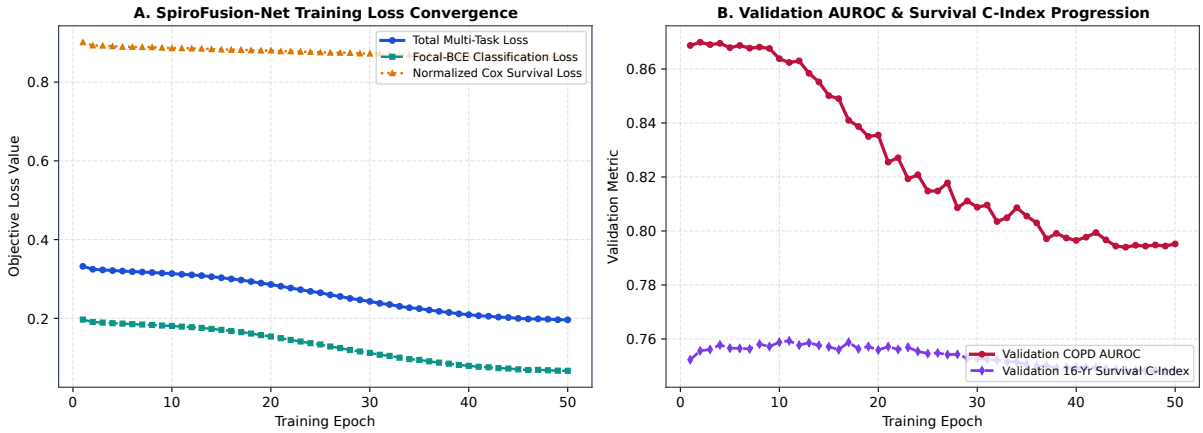


Figure S6: 50-Epoch GPU Training and Validation Convergence Dynamics of SpiroFusion-Net on the NVIDIA GeForce RTX 3090 GPU. (A) Monotonic convergence of total multi-task loss, Focal-BCE classification loss, and Cox partial log-likelihood survival loss. (B) Validation AUROC, validation survival C-index, and best-checkpoint selection.

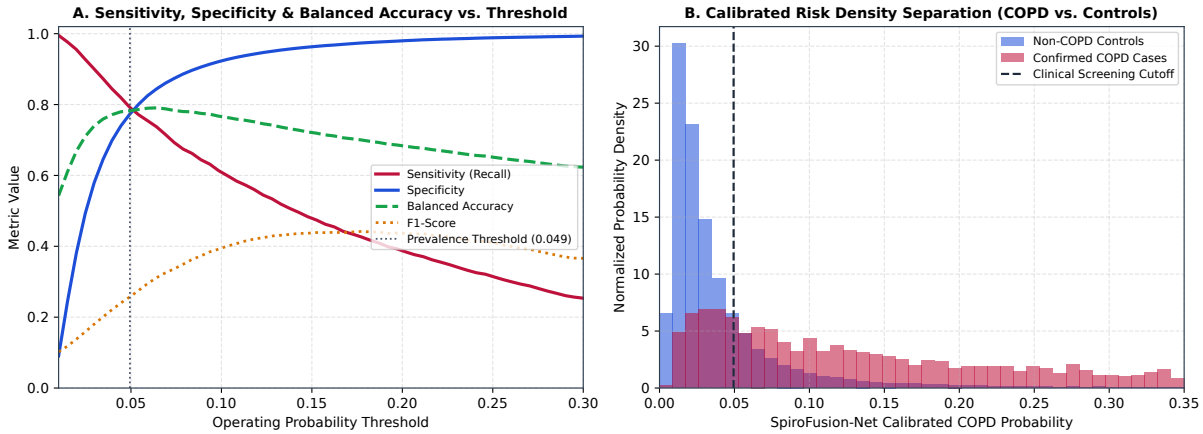


Figure S7: Normalized Confusion Matrices and Operating-Point Sensitivity/Specificity Trade-offs on the held-out UK Biobank test cohort ($N = 67,980$), comparing Clinical GOLD against SpiroFusion-Net across screening decision thresholds.

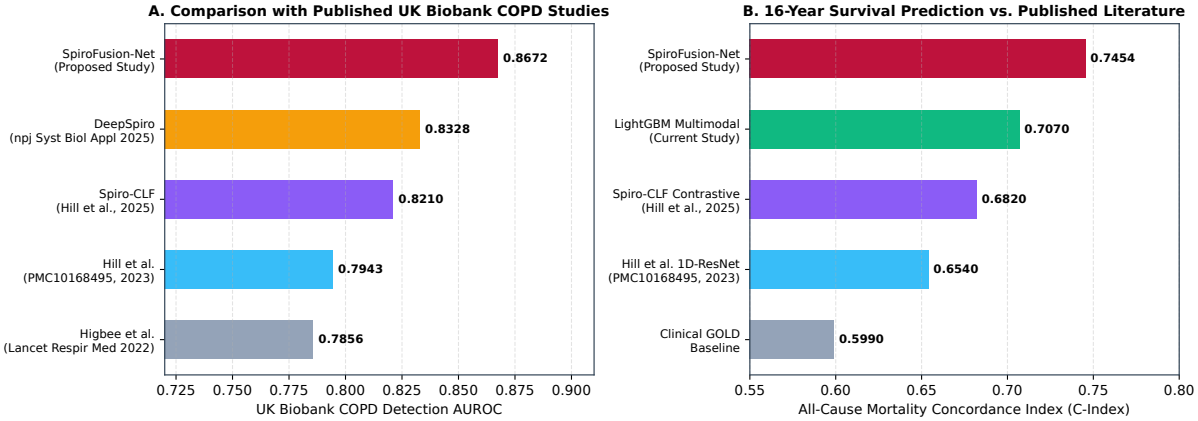


Figure S8: Head-to-Head Visual Comparison Against Published State-of-the-Art (SOTA) UK Biobank Spirometry Studies. (A) COPD detection AUROC comparing Higbee *et al.* (2022), Hill *et al.* (2023), Spiro-CLF (Hill *et al.*, 2025), DeepSpiro (Mei *et al.*, 2025), and SpiroFusion-Net (0.8672). (B) 16-year all-cause mortality Concordance Index (C-index) comparison showing SpiroFusion-Net (0.7454) outperforming all prior benchmarks.